Docker Challenges - 01

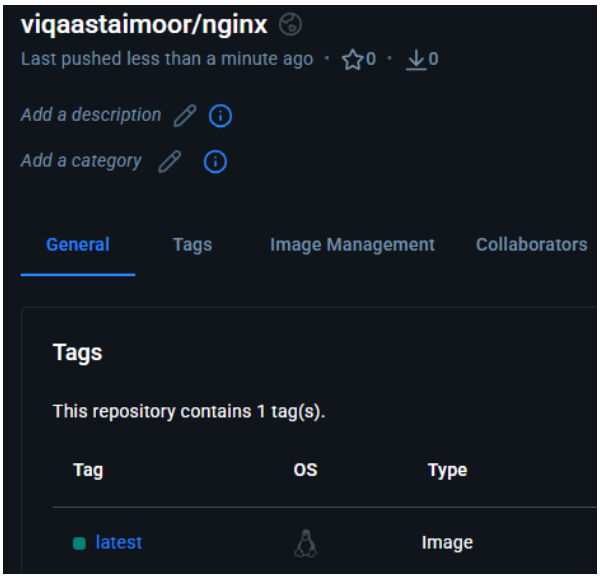
- Create a customized Docker image using a Dockerfile.

```
[root@ip-172-31-24-34 ~]# ls
Dockerfile  index.html
[root@ip-172-31-24-34 ~]# cat Dockerfile
FROM amazonlinux:latest

LABEL maintainer="V"

RUN dnf install -y nginx

COPY index.html /usr/share/nginx/html/

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

```
[root@ip-172-31-24-34 ~]# docker build -t test:v1 .
[+] Building 0.3s (9/9) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 258B
 => [internal] load metadata for docker.io/library/amazonlinux:latest
 => [auth] library/amazonlinux:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/amazonlinux:latest@sha256:162fc5b69e11e81023f83a1ab618b184927ff3002d0852432af3b6ae4a1b5304
 => [internal] load build context
 => => transferring context: 92B
 => CACHED [2/3] RUN dnf install -y nginx
 => CACHED [3/3] COPY index.html /usr/share/nginx/html/
 => exporting to image
 => => exporting layers
 => => writing image sha256:64048fae95632685c9647b0dd1c8fde7ae4256ac1e29c2a4f3aedba6a0e0bf09
 => => naming to docker.io/library/test:v1
```

- Push the image to Docker Hub.

```
[root@ip-172-31-24-34 ~]# docker tag test:v1 viqaastaimoor/nginx:latest
[root@ip-172-31-24-34 ~]# docker push viqaastaimoor/nginx:latest
The push refers to repository [docker.io/viqaastaimoor/nginx]
a50dba72c074: Mounted from viqaastaimoor/mynginx
18c0ad3a26ff: Mounted from viqaastaimoor/mynginx
4268ad241b85: Mounted from viqaastaimoor/mynginx
latest: digest: sha256:34831de79ae4b5625cd3c8979ce4a2b8def073259643fa3e1e64fa759c827c27 size: 949
[root@ip-172-31-24-34 ~]#
```



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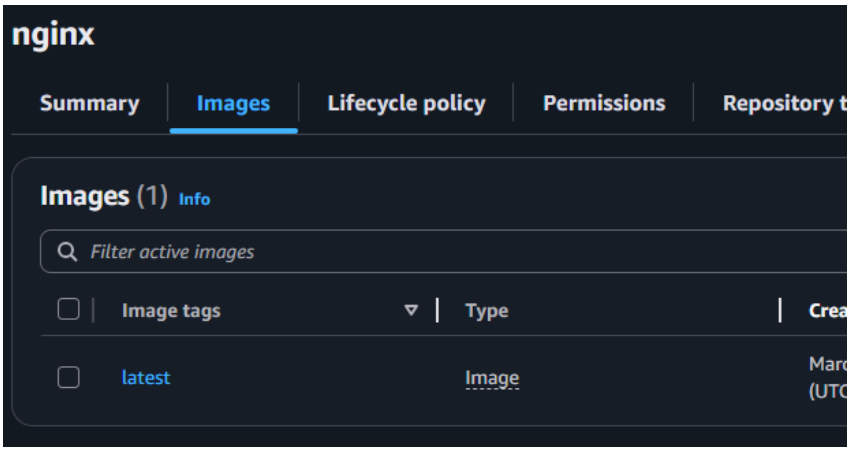
- Push the same image to Amazon ECR.

```
[root@ip-172-31-24-34 ~]# aws ecr create-repository --repository-name nginx
{
    "repository": {
        "repositoryArn": "arn:aws:ecr:us-east-1:526018540742:repository/nginx",
        "registryId": "526018540742",
        "repositoryName": "nginx",
        "repositoryUri": "526018540742.dkr.ecr.us-east-1.amazonaws.com/nginx",
        "createdAt": "2026-03-27T06:25:52.257000+00:00",
        "imageTagMutability": "MUTABLE",
        "imageScanningConfiguration": {
            "scanOnPush": false
        },
        "encryptionConfiguration": {
            "encryptionType": "AES256"
        }
    }
}
[root@ip-172-31-24-34 ~]# aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 526018540742.dkr.ecr.us-east-1.amazonaws.com
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-172-31-24-34 ~]# docker tag test:v1 526018540742.dkr.ecr.us-east-1.amazonaws.com/nginx:latest
[root@ip-172-31-24-34 ~]# docker push 526018540742.dkr.ecr.us-east-1.amazonaws.com/nginx:latest
The push refers to repository [526018540742.dkr.ecr.us-east-1.amazonaws.com/nginx]
a50dba72c074: Pushed
18c0ad3a26ff: Pushed
4268ad241b85: Pushed
latest: digest: sha256:34831de79ae4b5625cd3c8979ce4a2b8def073259643fa3e1e64fa759c827c27 size: 949
```



- Provision one EC2 instance using Terraform and install Jenkins.

Docker > main.tf > resource "aws_instance" "jenkins" > ami

```tf
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "jenkins" {
  ami           = "ami-0f4b77a23eac2378a"
  instance_type = "t2.micro"
  key_name      = "Linux-key"

  tags = {
    Name = "Jenkins-Server"
  }

  user_data = <<-EOF
              #!/bin/bash
              dnf install java-17-amazon-corretto -y
              wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
              rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
              dnf install jenkins -y
              systemctl enable jenkins
              systemctl start jenkins
              EOF
}
```

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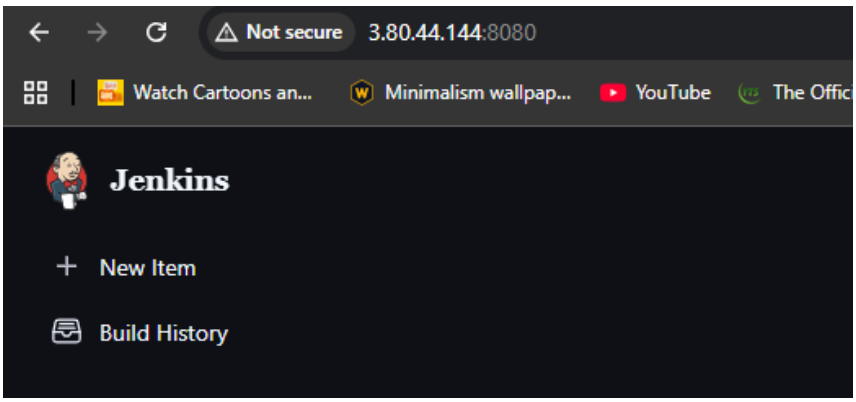
```
Plan: 1 to add, 0 to change, 0 to destroy.
aws_instance.jenkins: Creating...
aws_instance.jenkins: Still creating... [00m10s elapsed]
aws_instance.jenkins: Still creating... [00m20s elapsed]
aws_instance.jenkins: Still creating... [00m30s elapsed]
aws_instance.jenkins: Creation complete after 36s [id=i-05256d2012a5d3372]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\Docker>
```



- Create one Jenkins job to build and push the Docker image to Docker Hub.

```
[root@ip-172-31-24-34 Docker]# cat Dockerfile
FROM python:3.9-slim

WORKDIR /app
RUN echo "print('Hello from Jenkins Pipeline')" > app.py

CMD ["python", "app.py"]
[root@ip-172-31-24-34 Docker]#
```

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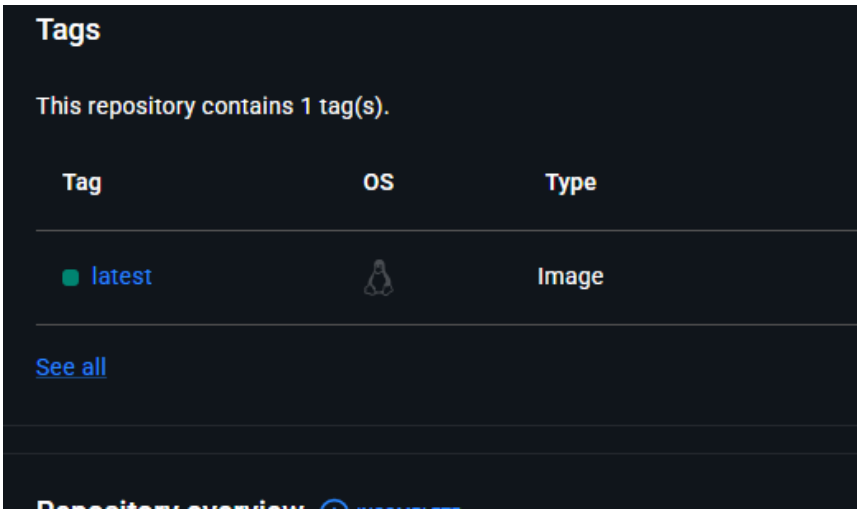
```
pipeline {
    agent any

    stages {
        stage('Build Image') {
            steps {
                sh 'docker build -t myapp .'
            }
        }

        stage('Login to Docker Hub') {
            steps {
                sh '''
                docker login -u viqaastaimoor -p dckr_pat_G-mg2GWwI7Oh2Cuj57BJ-KuwFQk
                '''
            }
        }

        stage('Tag Image') {
            steps {
                sh 'docker tag myapp viqaastaimoor/myapp:latest'
            }
        }

        stage('Push Image') {
            steps {
                sh 'docker push viqaastaimoor/myapp:latest'
            }
        }
    }
}
```



- **Source Codes:** https://github.com/betawins/docker-tasks.git
- From the **frontend source code**, write a Dockerfile, build a Docker image, run it, and push that image to your Docker registry.

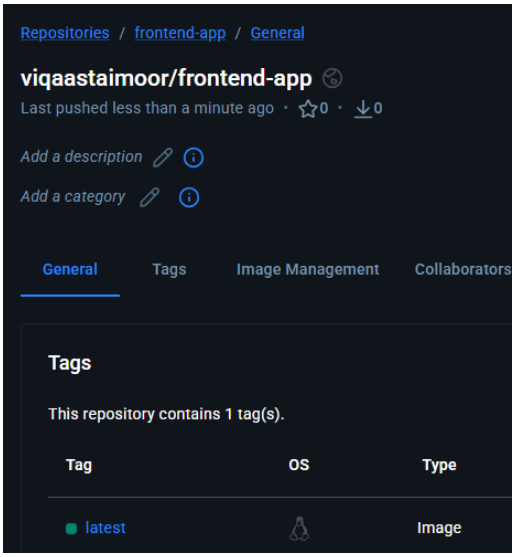
Docker Challenges - 01

```
[root@ip-172-31-24-34 docker-tasks]# ls
Frontend_based_source  Java_based_source  NodeJs_based_source  README.md
[root@ip-172-31-24-34 docker-tasks]# cd Frontend_based_source/
[root@ip-172-31-24-34 Frontend_based_source]# cat Dockerfile
FROM nginx:alpine

COPY . /usr/share/nginx/html
[root@ip-172-31-24-34 Frontend_based_source]# docker build -t viqaastaimoor/frontend-app .
[+] Building 1.9s (8/8) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 145B
 => [internal] load metadata for docker.io/library/nginx:alpine
 => [auth] library/nginx:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
```

```
[root@ip-172-31-24-34 Frontend_based_source]# docker run -d -p 8085:80 --name frontend viqaastaimoor/frontend-app
55d2142692f3a3c6d397e4be69c4f5682b6e1642507702abbea4d5b4a8b9c3f6
[root@ip-172-31-24-34 Frontend_based_source]# docker ps
CONTAINER ID   IMAGE                        COMMAND                  CREATED         STATUS         PORTS                                      NAMES
55d2142692f3   viqaastaimoor/frontend-app   "/docker-entrypoint.…"   3 seconds ago   Up 2 seconds   0.0.0.0:8085->80/tcp, :::8085->80/tcp   frontend
[root@ip-172-31-24-34 Frontend_based_source]#
```

```
[root@ip-172-31-24-34 Frontend_based_source]# docker push viqaastaimoor/frontend-app
Using default tag: latest
The push refers to repository [docker.io/viqaastaimoor/frontend-app]
7cb9eff6ace8: Pushed
6d895dca2a49: Mounted from library/nginx
e8abceee8fd8: Mounted from library/nginx
1eb65b5fb2a1: Mounted from library/nginx
cff742fa7408: Mounted from library/nginx
9ffd725bd2f3: Mounted from library/nginx
8ca391b8823c: Mounted from library/nginx
27fc2bd29afd: Mounted from library/nginx
989e799e6349: Mounted from library/nginx
latest: digest: sha256:6bce107735ff564f3eb8384fbc4cafab5fe45bb9c3ee1a876b3c7c4d223c51fd size: 2197
[root@ip-172-31-24-34 Frontend_based_source]#
```



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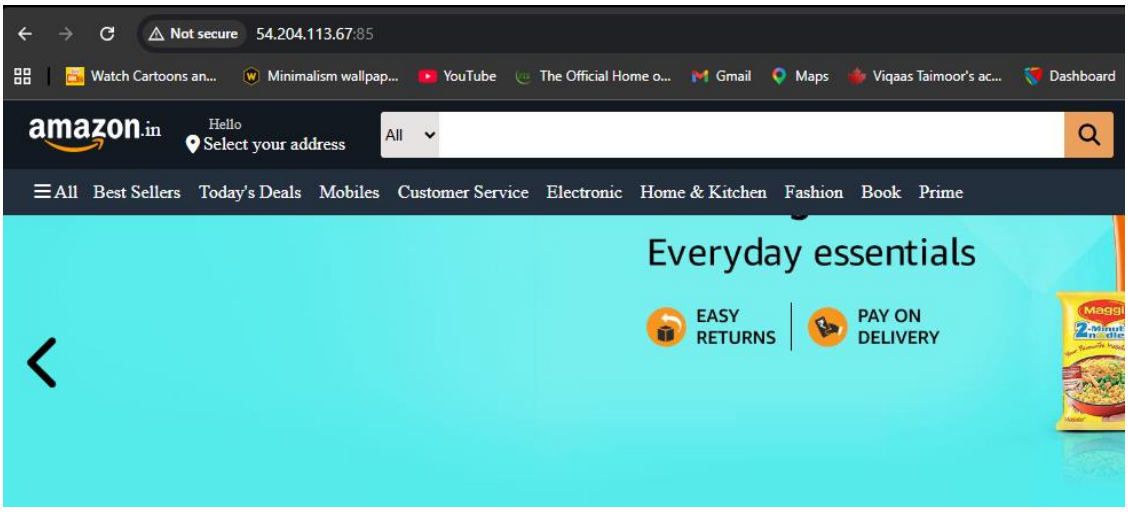

- From the **Java-based source code**, write a Dockerfile, build, run, and push the image to the Docker registry.

```
[root@ip-172-31-24-34 Java_based_source]# ls
Dockerfile  pom.xml  src
[root@ip-172-31-24-34 Java_based_source]# cat Dockerfile
FROM maven:3.8.6-openjdk-11 AS build

WORKDIR /app
COPY . .

RUN mvn clean package

FROM eclipse-temurin:11-jre

COPY --from=build /app/target/*.jar app.jar

CMD ["java", "-jar", "app.jar"]
[root@ip-172-31-24-34 Java_based_source]# docker build -t viqaastaimoor/java:latest .
[+] Building 26.1s (14/14) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 289B
 => [internal] load metadata for docker.io/library/eclipse-temurin:11-jre
 => [internal] load metadata for docker.io/library/maven:3.8.6-openjdk-11
 => [auth] library/maven:pull token for registry-1.docker.io
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [build 1/4] FROM docker.io/library/maven:3.8.6-openjdk-11@sha256:805f366910aea2a91ed26
 => [internal] load build context
 => => transferring context: 1.89MB
 => CACHED [stage-1 1/2] FROM docker.io/library/eclipse-temurin:11-jre@sha256:52c91c17a870
```

```
[root@ip-172-31-24-34 Java_based_source]# docker run -d -p 8085:8080 --name java viqaastaimoor/java:latest
c6f975d79b640335157e7e76936ed19bb9f615d178feed63a808b8bd5899566f
[root@ip-172-31-24-34 Java_based_source]# docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED         STATUS         PORTS                                       NAMES
c6f975d79b64   viqaastaimoor/java:latest  "/__cacert_entrypoin…"   2 seconds ago   Up 2 seconds   0.0.0.0:8085→8080/tcp, :::8085→8080/tcp   java
[root@ip-172-31-24-34 Java_based_source]#
```

```
[root@ip-172-31-24-34 Java_based_source]# docker push viqaastaimoor/java:latest
The push refers to repository [docker.io/viqaastaimoor/java]
6166125bcbed: Pushed
61a6f8affb50: Mounted from library/eclipse-temurin
56437b53119b: Mounted from library/eclipse-temurin
ec6e2d3734df: Mounted from library/eclipse-temurin
c4e3850aa7d4: Mounted from library/eclipse-temurin
f2a7f0726353: Mounted from library/eclipse-temurin
latest: digest: sha256:8d9889dd8c044d29dcf798ada04a29df66153a88673a9d64b67576b7549d1f57 size: 1580
[root@ip-172-31-24-34 Java_based_source]#
```

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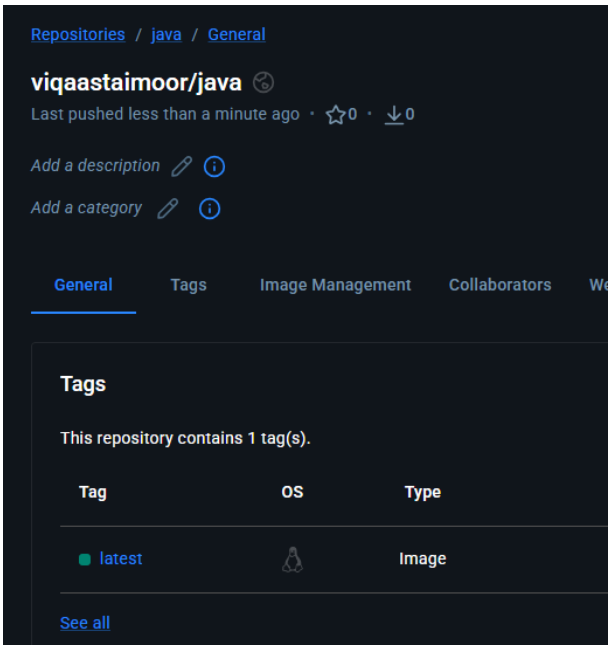

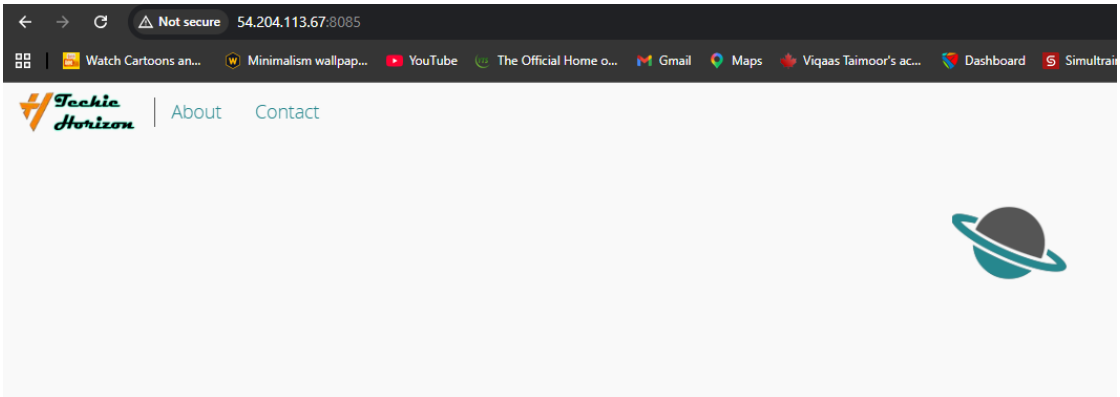

- From the **Node.js-based source code**, write a Dockerfile, build with tag `v1`, run, and push it to the Docker registry.

```
[root@ip-172-31-24-34 NodeJs_based_source]# ls
Dockerfile  package-lock.json  package.json  public  src
[root@ip-172-31-24-34 NodeJs_based_source]# cat Dockerfile
FROM node:18

WORKDIR /app
COPY . .

RUN npm install

CMD ["npm", "start"]
[root@ip-172-31-24-34 NodeJs_based_source]# docker build -t viqaastaimoor/node:v1 .
[+] Building 84.2s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 172B
 => [internal] load metadata for docker.io/library/node:18
 => [auth] library/node:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/4] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96
```

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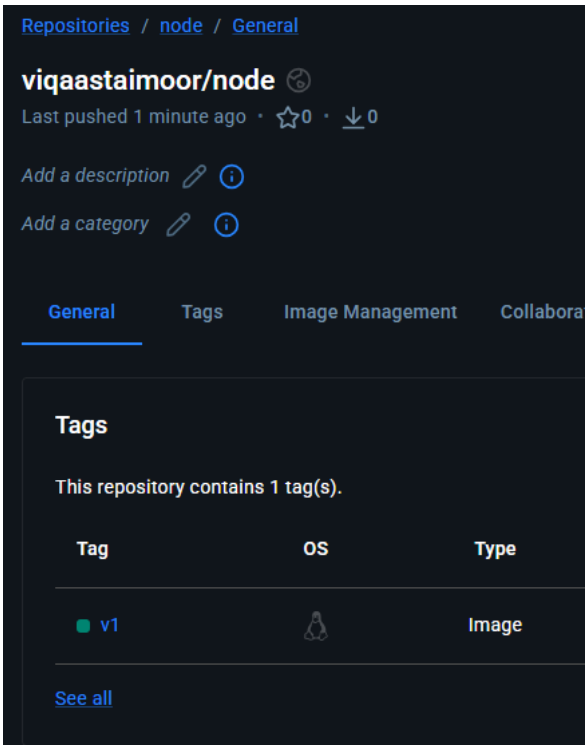
```
[root@ip-172-31-24-34 NodeJs_based_source]# docker run -d -p 3000:3000 --name node viqaastaimoor/node:v1
20bcf0d525a9cfa62d2d512adf4ef5caa949cdcd0ea41b0b9f0ca74eeb4676f1
[root@ip-172-31-24-34 NodeJs_based_source]# docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS         PORTS                                       NAMES
20bcf0d525a9   viqaastaimoor/node:v1  "docker-entrypoint.s…"   3 seconds ago   Up 2 seconds   0.0.0.0:3000→3000/tcp, :::3000→3000/tcp   node
[root@ip-172-31-24-34 NodeJs_based_source]#
```

```
[root@ip-172-31-24-34 NodeJs_based_source]# docker push viqaastaimoor/node:v1
The push refers to repository [docker.io/viqaastaimoor/node]
50786f3d9127: Pushed
419ad21c612e: Pushed
ac349c3bc720: Pushed
d2a991bcab4d: Mounted from library/node
b624aa2d5ea2: Mounted from library/node
d399c9dc306f: Mounted from library/node
84f9fa179c1b: Mounted from library/node
ce84ba212e49: Mounted from library/node
e4dc8cd9ecc8: Mounted from library/node
6428cc293366: Mounted from library/node
2f7436e79a0b: Mounted from library/node
v1: digest: sha256:7ffc21d1b2817f73a787ba445971fa7e6e12b7cddc57dfd5d4f259b73f259452 size: 2634
[root@ip-172-31-24-34 NodeJs_based_source]#
```



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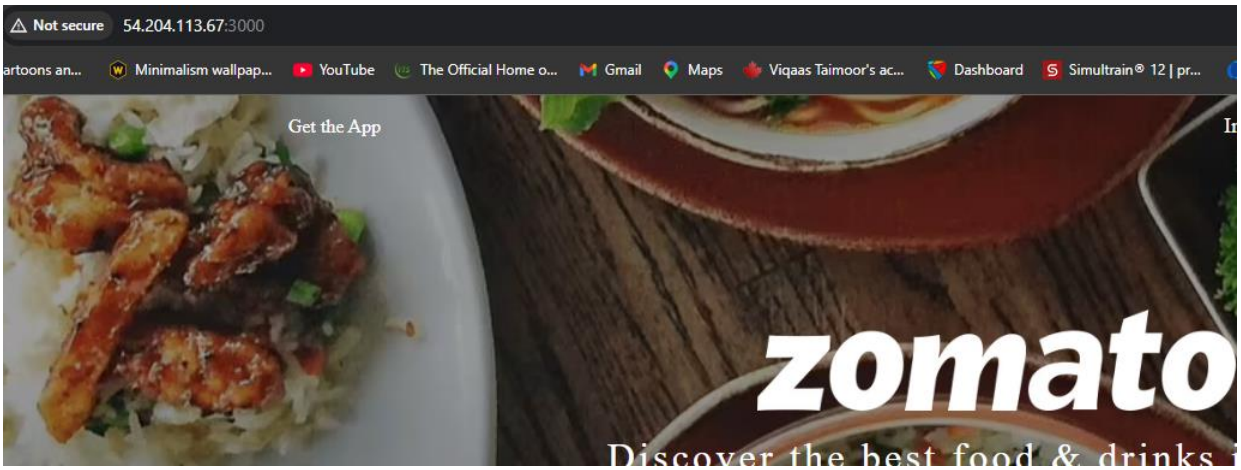

- Write a **docker-compose file** to set up WordPress with a MySQL database.

```
[root@ip-172-31-24-34 ~]# cat docker-compose.yml
services:
  db:
    image: mysql:5.7
    container_name: mysql
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress
    restart: always
    ports:
      - "8081:80"
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    depends_on:
      - db

volumes:
  db_data:
```

```
[root@ip-172-31-24-34 ~]# docker compose up -d
[+] up 2/2
 ✔ Container mysql     Running
 ✔ Container wordpress Started
[root@ip-172-31-24-34 ~]#
```

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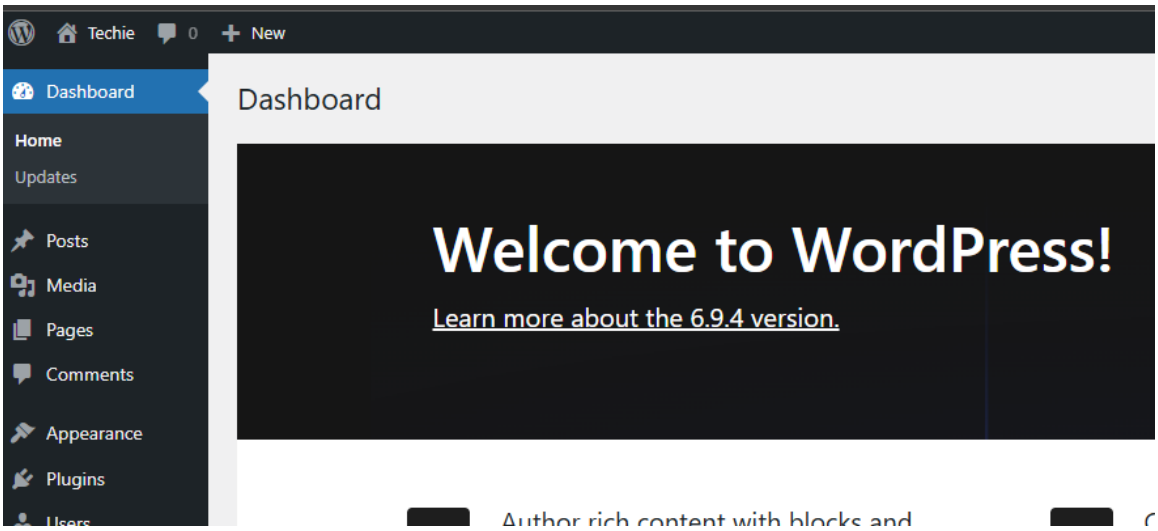


```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 25
Server version: 5.7.44 MySQL Community Server (GPL)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| wordpress          |
+--------------------+
5 rows in set (0.00 sec)
```